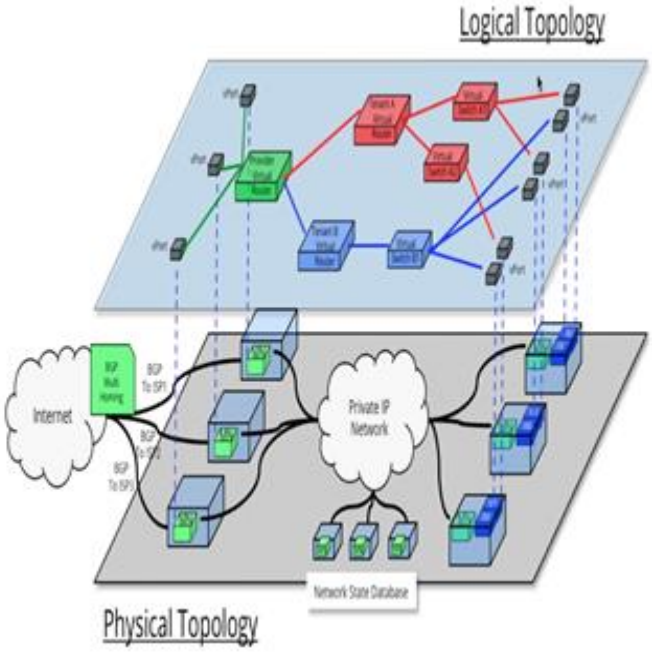


CLOUD COMPUTING COURSE

Lecture13-Network Virtualization in CC (3)

ABSTRACT

Lecture14 Introduces network topologies and network types

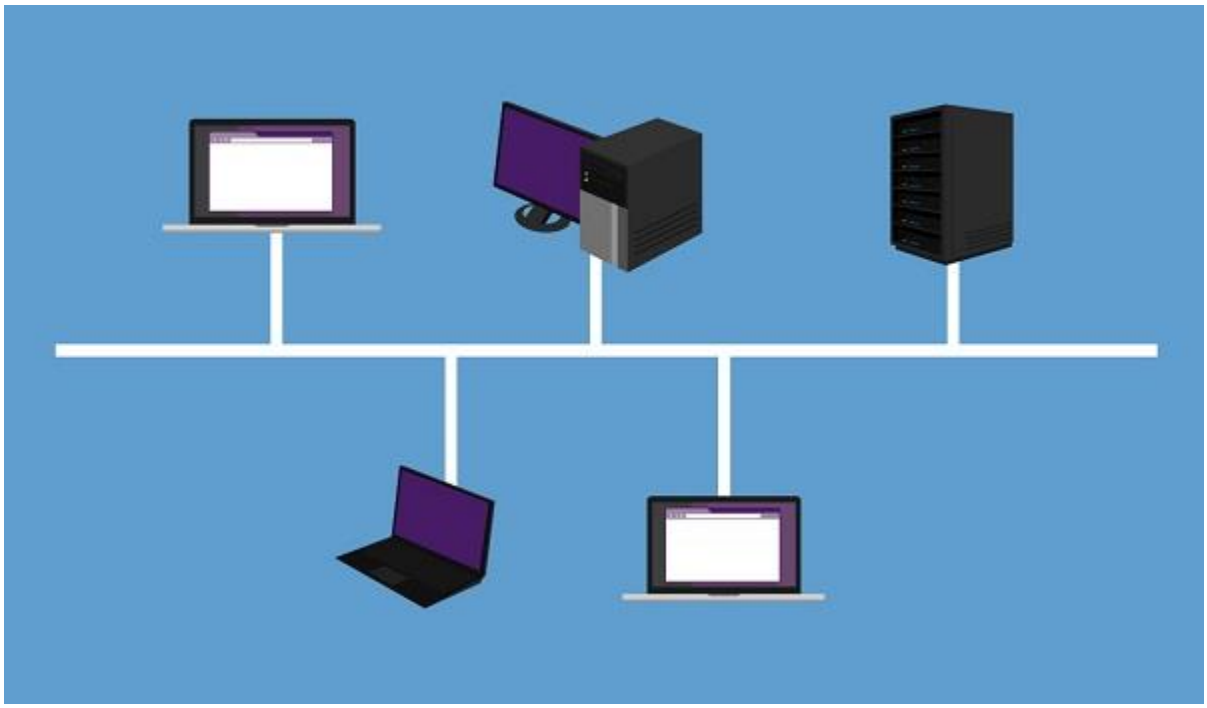
Instructor: Asma'a Khtoom

1. Network Topologies

The way in which devices are interconnected to form a network is called network topology.

1. Bus Topology

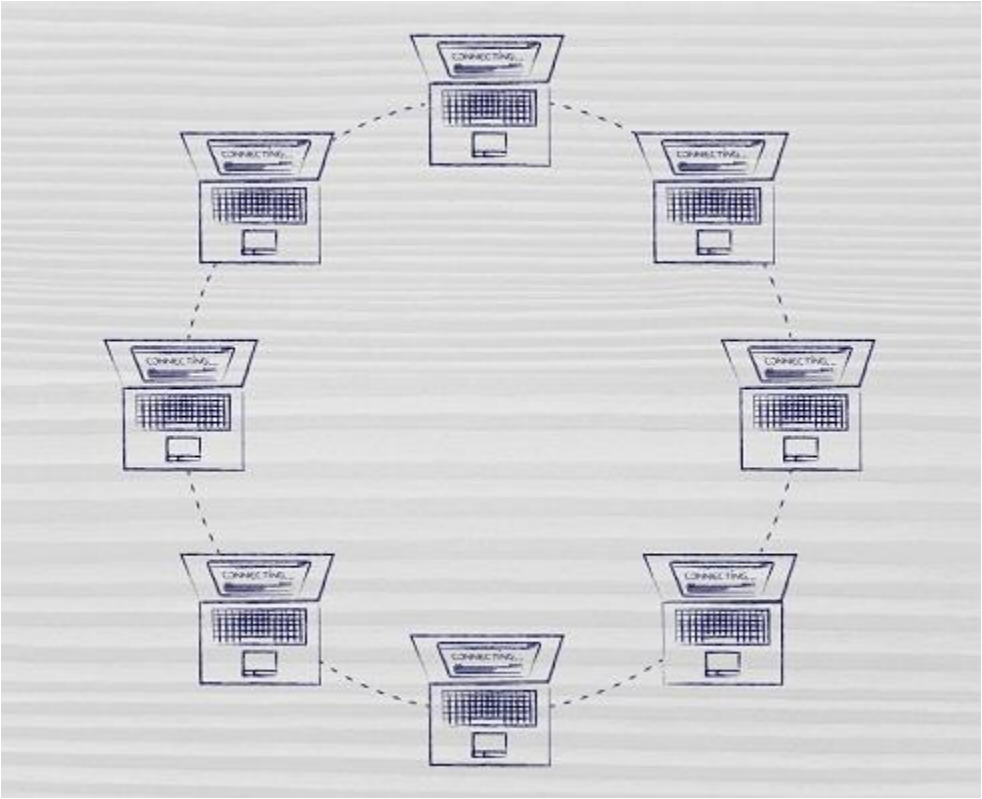
Data network with bus topology has a **linear transmission cable**, usually **coaxial**, to which many **network devices** and **workstations** are attached along the length. **Server** is at one end of the bus. When a workstation has to send data, it transmits **packets** with **destination address** in its header along the bus.



The data travels in both the directions along the bus. When the destination terminal sees the data, it copies it to the local disk.

2. Ring Topology

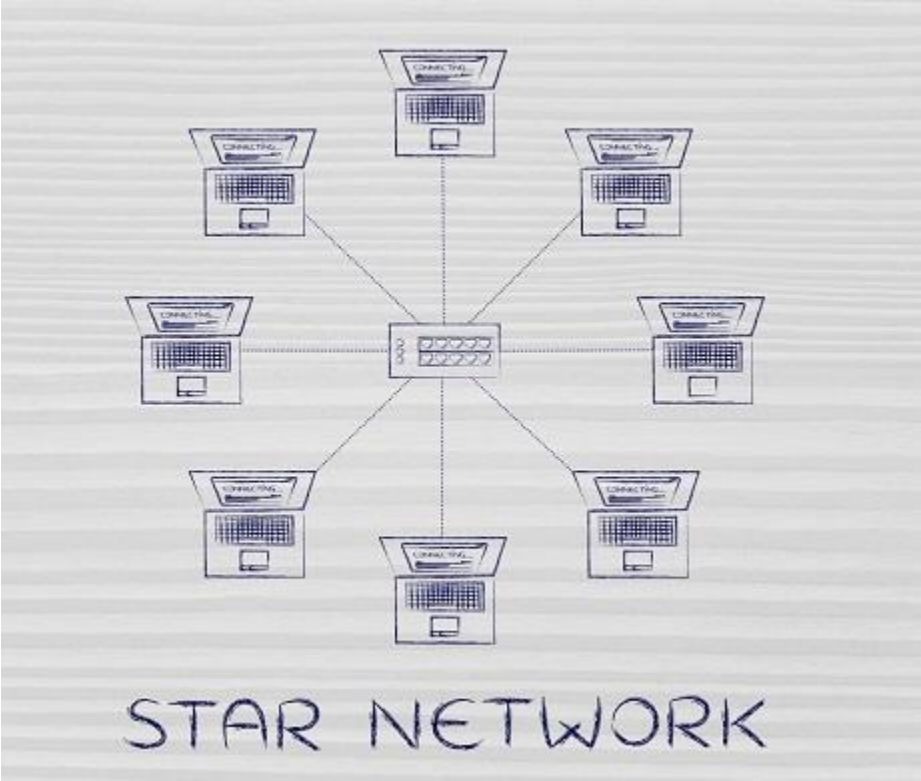
In **ring topology** each terminal is connected to exactly **two nodes**, giving the network a circular shape. Data travels in only one pre-determined direction.



When a terminal has to send data, it transmits it to the neighboring node which transmits it to the next one. Before further transmission data may be amplified. In this way, data raverses the network and reaches the destination node, which removes it from the network. If the data reaches the sender, it removes the data and resends it later.

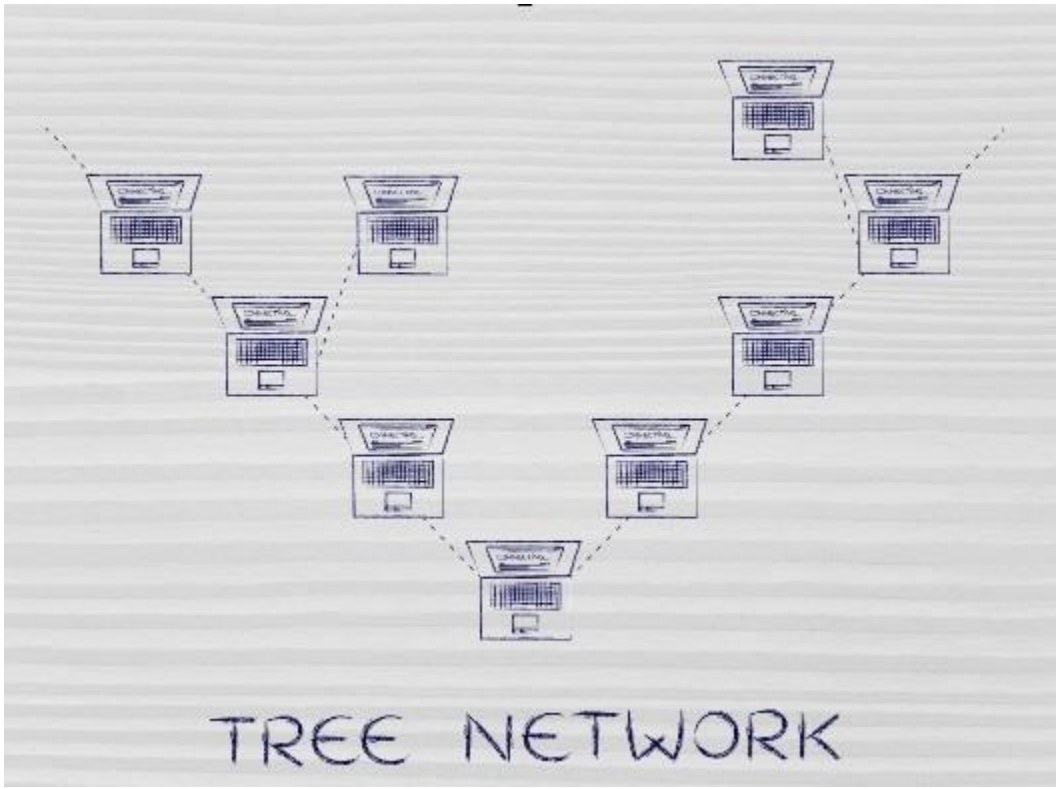
3. Star Topology

In star topology, server is connected to each node individually. Server is also called the central node. Any exchange of data between two nodes must take place through the server. It is the most popular topology for information and voice networks as central node can process data received from source node before sending it to the destination node.



4. Tree Topology

Tree topology has a group of star networks connected to a linear bus backbone cable. It incorporates features of both star and bus topologies. Tree topology is also called hierarchical topology.

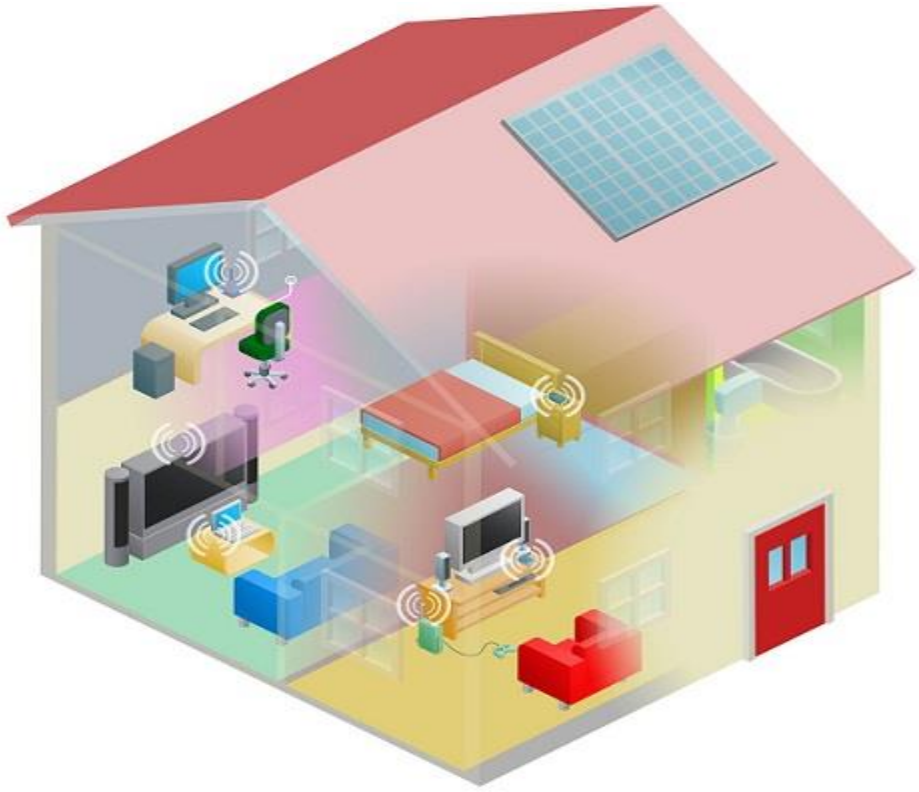


2. Network Types

Networks can be categorized depending on size, complexity, level of security, or geographical range. We will discuss some of the most popular topologies based on geographical spread.

1. PAN

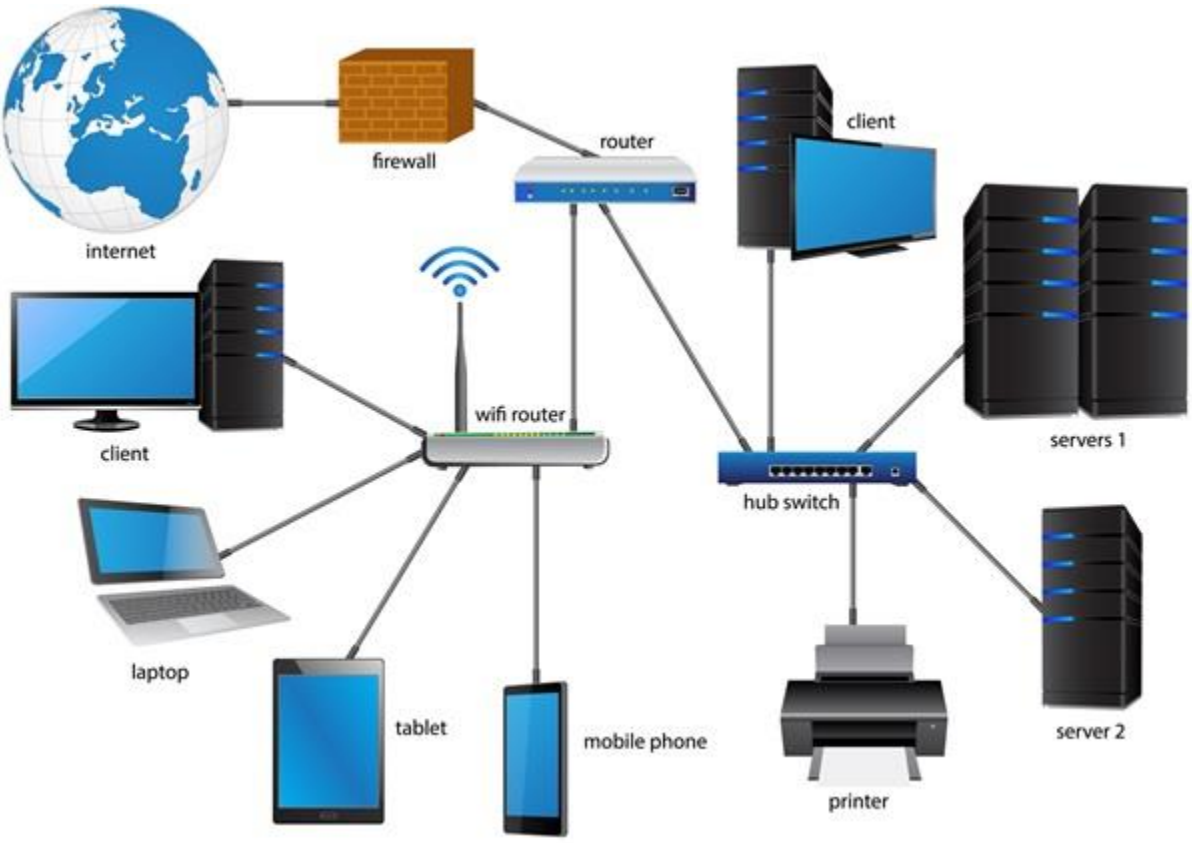
PAN is the acronym for Personal Area Network. PAN is the interconnection between devices within the range of a person's private space, typically within a range of 10 metres. If you have transferred images or songs from your laptop to mobile or from mobile to your friend's mobile using Bluetooth, you have set up and used a personal area network.



A person can connect her laptop, smart phone, personal digital assistant and portable printer in a network at home. This network could be fully Wi-Fi or a combination of wired and wireless.

2. LAN

LAN or Local Area Network is a wired network spread over a single site like an office, building or manufacturing unit. LAN is set up to when team members need to share software and hardware resources with each other but not with the outside world. Typical software resources include official documents, user manuals, employee handbook, etc. Hardware resources that can be easily shared over the network include printer, fax machines, modems, memory space, etc. This decreases infrastructure costs for the organization drastically.



A LAN may be set up using wired or wireless connections. A LAN that is completely wireless is called Wireless LAN or WLAN.

3. MAN

MAN is the acronym for Metropolitan Area Network. It is a network spread over a city, college campus or a small region. MAN is larger than a LAN and typically spread over several kilometres. Objective of MAN is to share hardware and software resources, thereby decreasing infrastructure costs. MAN can be built by connecting several LANs.



The most common example of MAN is cable TV network.

4. WAN

WAN or Wide Area Network is spread over a country or many countries. WAN is typically a network of many LANs, MANs and WANs. Network is set up using wired or wireless connections, depending on availability and reliability.



The most common example of WAN is the **Internet.**

Related Video: Please watch the next videos:

1. **Network Topologies:** ***https://youtu.be/zbqrNg4C98U***

2. **Network Types:** ***https://youtu.be/4_zSIXb7tLQ***